



# AFFINITY™ PL 1881G

## Polyolefin Plastomer

AFFINITY™ PL 1881G Polyolefin Plastomer is a Polyolefin Plastomer (POP) produced using INSITE™ Technology from Dow Plastics<sup>†</sup>. It is specifically designed for use as a sealant layer in flexible structures for the packaging of meat and cheese, dry foods and consumer goods. Due to its excellent sealability at low temperatures, ultimate hot tack strength, optical properties and abuse resistance, it is particularly suitable for high speed form-fill-seal packaging machines. This resin contains slip and antiblocking additives

**Note:** When used unmodified for food contact applications, this resin will comply with the Federal Food, Drug, and Cosmetic Act as a food contact substance (FCN) as a result of a premarket notification with an effective date of October 7, 2004 under FCN 424. This notification allows for use of this product as articles or components of articles used in contact with all food types under Conditions of Use A through H, as described in U.S. FDA Table 2 of 21 CFR § 176.170(c). The composition of this product complies with the requirements for use in contact with food of the EU-Directive 2002/72/EC. Please contact your nearest Dow office regarding food contact compliance statements. The purchaser remains responsible for determining whether the use complies with all relevant regulations.

| Physical Properties <sup>(1)</sup>                | Unit              | Test Method | Values |
|---------------------------------------------------|-------------------|-------------|--------|
| Melt Index, 190 °C/2.16 kg                        | g/10min           | ISO 1133    | 1      |
| Density                                           | g/cm <sup>3</sup> | ASTM D-792  | 0.904  |
| Melting Point                                     | °C                | DSC         | 100    |
| Vicat Softening Point                             | °C                | ASTM D-1525 | 88     |
| Film Properties, 50 µm thickness <sup>(1,2)</sup> | Unit              | Test Method | Values |
| Puncture Resistance                               |                   | ASTM D-5748 |        |
| Energy                                            | J                 |             | 1.6    |
| Force                                             | N                 |             | 30     |
| Dart Impact, (Method A)                           | g                 | ISO 7765-1  | >2600  |
| Elmendorf Tear                                    | g                 | MD          | 580    |
|                                                   |                   | CD          | 770    |
| Tensile Yield                                     | MPa               | MD          | 6.1    |
|                                                   |                   | CD          | 6.2    |
| Ultimate Tensile                                  | MPa               | MD          | 44     |
|                                                   |                   | CD          | 47     |
| Ultimate Elongation                               | %                 | MD          | 610    |
|                                                   |                   | CD          | 590    |
| Tensile Modulus, 2% Secant                        | MPa               | MD          | 70     |
|                                                   |                   | CD          | 70     |
| Optical Properties                                |                   |             |        |
| Gloss 20°                                         | units             | ASTM D-2457 | 105    |
| Haze                                              | %                 | ISO 14782   | 3.6    |
| Seal Initiation Temperature <sup>(3)</sup>        | °C                | Dow Method  | 89     |

(1) Typical values, not to be construed as specifications limits.

(2) Monolayer blown film, extruded at 221 °C, BUR 2.5:1, 1.8 mm die gap.

(3) Temperature required to reach 5.25 N/15 mm heat-seal strength.

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## Additional Information

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